



LAKSHMINARASIMHAN R

B.Tech Electronics and Communication Engineering

National Institute of Technology Calicut

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Roll No: B240869EC

Education

- **National Institute of Technology Calicut** 2023 - Present
B.Tech Electronics and Communication Engineering CGPA: 7.09
- **St. Patrick Matric Higher Secondary School** 2022
Higher Secondary Education 82%
- **St. Patrick Matric Higher Secondary School** 2020
Secondary Education 100%

Internship

- **UAS Drone AI Internship**
Mechanical Engineering Department, NIT Calicut
- AI / Computer Vision Intern working on drone-based perception systems.
- Developed object detection models using YOLO and RCNN architectures.
- Created custom datasets including image collection, annotation and preprocessing.
- Trained and evaluated deep learning models for real-time detection.
- Implemented edge AI deployment using Raspberry Pi for drone applications.

Projects

- **Gesture Controlled Mouse using ESP32**
Wireless human-computer interaction system using ESP32 and MPU6050 accelerometer.
Technologies: ESP32, MPU6050, Arduino IDE, Embedded C.
- **Smart Blind Assistance System**
Assistive device for visually impaired users using ultrasonic obstacle detection.
Technologies: ESP32, Ultrasonic Sensor, Embedded C.
- **YOLO & RCNN Helmet Detection System**
Computer vision based safety monitoring system.
Technologies: YOLO, RCNN, Python, OpenCV, Deep Learning.

Experience

- **IEEE YESS** 2025
On-Ground Event Operations
- Managed event logistics, participant support and coordination activities.
- Worked with teams to ensure smooth execution of technical events.

- Pickngo** 2025
Marketing Intern
- Supported marketing activities and customer engagement strategies.
- Created promotional content and contributed to improving outreach.

NextgenOps 2026
Technical team
 Supported the team in AI ML domain.
 Created AI ML content and contributed to improving Website.

Technical Skills

Programming Languages: C, C++, Python, MATLAB

AI / Machine Learning: Machine Learning, Deep Learning, Computer Vision, YOLO, RCNN, OpenCV, Object Detection, Dataset Annotation

Embedded Systems: ESP32, Arduino, Raspberry Pi, Embedded C, Microcontrollers, Sensor Interfacing, IoT

ECE Tools: MATLAB, Cadence, LTspice, Arduino IDE, Keil uVision, 8086 Compiler

Core Electronics: Analog Electronics, Digital Electronics, Signal Processing, Communication Systems, Network Analysis, Control Systems, VLSI Fundamentals

Engineering Skills: Hardware-Software Integration, Debugging, Rapid Prototyping, Technical Documentation

Areas of Interest: Artificial Intelligence, Computer Vision, Robotics, Drone Technology(UAS), Embedded Systems, IoT, VLSI Design, Wireless Communication

Positions of Responsibility

- IEEE CASS** 2025
IEEE EXECom Member
- ECEA** 2025
Marketing Core Member

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